

Smartphone-Based 3D Visualization for Grape Berry Thinning Assistance –Supplementary Material–

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A Accuracy of Reconstruction and Berry Detection

This section presents detailed berry-detection accuracy data that could not be included in the main paper due to page limits. Table 1 shows the berry counts for the eight bunches for which detection succeeded. To assess the agreement between the detected counts and the ground-truth counts, Fig. 1 shows a Bland–Altman plot. The mean difference (bias) is -0.62 berries, close to zero, and the points scatter on both sides of this line. The 95% limits of agreement range from -4.54 to 3.29 berries, meaning that an individual bunch’s detected count can differ from its true count by up to about 4–5 berries. Bunch 3 is a clear outlier, with a difference of -4 berries lying near the lower limit of agreement. There is no clear trend across bunch size, though the three largest bunches (mean count above 70) all show a negative difference.

Table 1: Berry-count detection accuracy on eight Shine Muscat bunches. “|Err.|” is the absolute error between the detected and true counts.

Bunch	1	2	3	4	5	6	7	8
True count	72	76	56	59	52	67	49	78
Detected count	70	75	52	61	53	67	50	76
Err.	2	1	4	2	1	0	1	2

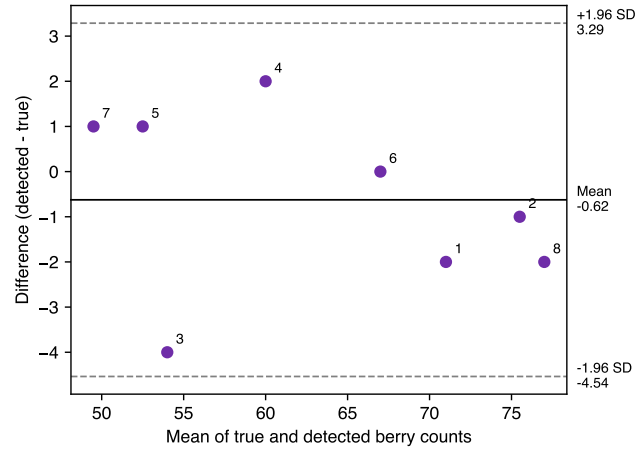


Fig. 1: Bland–Altman plot comparing detected and true berry counts for the eight bunches in Tab. 1. The solid line shows the mean difference (-0.62); dashed lines show the 95% limits of agreement (-4.54 , 3.29).